

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0016690483113057867
rotor delay [s]: 0.197554526304
gimbal delay [s]: 0.041459452946
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 4.747647607694383e-17
common-scale FD null norm: 1.2734061213338051e-11

Dimensionless coordinate

log_mass_scale	-0.161348957486
log_second_moment_mode_xx	-2.0749231183
log_second_moment_mode_yy	0.59670154075
log_second_moment_mode_zz	2.93246446972
log_second_moment_mode_xy	-0.0377312269878
log_second_moment_mode_xz	2.008173978
log_second_moment_mode_yz	0.00259598792006
dimensionless_cog_displacement_x	-0.0458390551821
dimensionless_cog_displacement_y	0.0151237740999
dimensionless_cog_displacement_z	-0.200796348943
log_force_effectiveness_1	-0.295004540617
log_force_effectiveness_2	-0.308936495109
log_force_effectiveness_3	-0.378999071511
log_force_effectiveness_4	-0.309953827444

Physical representative

mass [kg]: 2.0011979286369868
CoG body [m]: [-0.01080678 0.00286696 -0.02895982]
force effectiveness: [0.74452821 0.7342274 0.68454625 0.73348082]
inertia [kg m²]:
[0.129442377, 0.001707080882, -0.03729306322]
[0.001707080882, 0.1318011128, 0.0002330617453]
[-0.03729306322, 0.0002330617453, 0.2367553867]
principal moments [kg m²]: [0.11755271 0.13200292 0.24844325]

Per-bag raw residual-wrench RMS

failure_1_single: [0.43274253 0.22240366 0.83456377 0.11368644 0.13997957 0.15167106]